

**AMENDMENT NO. 1 FEBRUARY 2007
TO
IS 13160 : 1991 ALEURITIC ACID — SPECIFICATION**

(Page 1, *clause 4*) — Substitute the following for the existing:

'4 REQUIREMENTS

4.1 Mandatory Requirements

The material shall be off white in colour and shall comply with the requirements as given in Table 1.

4.2 Optional Requirements

Purity level of the material shall be as mutually agreed to by the purchaser and the supplier as ascertained by DSC analysis as outlined in A-7.'

(Page 2, *Annex A*) — Insert the following at the end:

A-7 TESTING OF PURITY USING DIFFERENTIAL SCANNING CALORIMETER (DSC)

A-7.1 General

The purity of aleuritic acid (9, 10, 16 trihydroxyhexadecanoic acid) can be determined from its melting profile (a plot of heat flow vs. temperature) using a Differential Scanning Calorimeter. The information about the above is obtained by analyzing the melting profile in a computer through appropriate software. A DSC also provides information about the melting temperature, a graphical presentation of the melting peak of the acid and presence of impurity, if any. The melting profile provides instantaneous information about the relative homogeneity/crystallinity of the material.

A-7.2 Apparatus

- a) A Differential Scanning Calorimeter with different rates of heating, computer controlled, with software for peak position, selection of ranges of dependent and independent variables, onset temperature, purity, percentage solid, partial area, etc.

The temperature range required for the analysis of aleuritic acid is 40-150°C.

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- b) Printer, printing papers for the output from DSC analysis*
- c) Aluminium pans with cover for DSC analysis with crimper*
- d) A good weighing balance with accuracy of ± 0.01 mg.

*The equipment/material would normally be provided with DSC.

A-7.3 Procedure

A-7.3.1 Weigh finely powdered 5-6 mg of aleuritic acid, accurately, in the aluminium pan, put the cover and crimp properly.

A-7.3.2 Switch on the printer, DSC (precalibrated) and controller. Start purging the sample holder with Ar or N₂. Start the compressor/circulator.

A-7.3.3 Place the sample in the crimped pan in the sample holder of DSC.

A-7.3.4 Start the software, select the appropriate menu for a temperature profile run. Enter the sample, file name, etc. Enter start and end temperatures, run rate (20°C/min), etc.

A-7.3.5 After attaining the steady state, start run, save the profile after completion of the run.

A-7.3.6 Enter into analysis software, select the appropriate portion of the profile, preferably containing the initial and final plateau regions.

A-7.4 Examination of the Melting Profile

Pure crystalline aleuritic acid gives sharp single melting peak. Presence of impurity gives broad peak associated with smaller peaks, shoulders, etc. Broader peak is indicative of heterogeneity.

The onset temperatures for threo and erythroisomers of aleuritic acid were observed to be 94.9 and 119.3°C, respectively. The data are based on samples crystallized in water.

A-7.5 Analysis of the Profile

A-7.5.1 *Melting Temperature*

Position of the melting peak will be shown on the screen, which can be taken as melting temperature. In case of good crystalline material, melting peak will be obtained as mentioned in **A-7.4**. Melting temperature will depend on method of

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crystallization. In case broad melting peak or more than one peak is obtained, further crystallization of the sample is to be done till sharp single peak is obtained.

A-7.5.2 Purity

Open the Purity menu. Enter the weight of the sample taken and its molecular weight. Select proper regions. Enter the purity calculation. Repeat the operation for different regions. Select purity values from a set of near similar values.

A-7.6 Report

The report shall contain position of the peak (melting temperature) in degrees Celsius, purity (%), for multiple peaks, the number of peaks with their positions. The sample shall be considered passing if the nature of peak and purity level is as per those agreed to, mutually, by the purchaser and the supplier.

(CHD 23)

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